
Intel Arc Pro B60

Platform Reference — Specification, Memory, Software Stack
and Compute Throughput by Datatype

Xe2-HPG “Battlemage” · BMG-G21 · 24 GB GDDR6
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GenAI / Agentic AI Infrastructure Reference Library

Provenance tags used throughout:

[I] Intel-published · **[D]** Derived, formula shown · **[3P]** Third-party · **[U]** Explicitly unpublished

No untraceable figure appears in a specification table.

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0. Document control and provenance discipline

Every quantitative claim in this document carries a provenance tag. Untagged prose is editorial framing, not data.

Tag	Meaning
[I]	Intel-published. Traceable to an Intel data sheet, product page, developer article or Intel architecture disclosure.
[D]	Derived. Computed from [I] inputs; the formula is shown inline so the arithmetic can be re-checked.
[3P]	Third-party. Board partner data sheet, press disclosure or reviewer measurement. Not confirmed by Intel.
[U]	Unpublished. Explicitly called out as <i>not available</i> so it is not silently filled in with a guess.

Two rules were applied while compiling this document:

1. **No unverifiable figure is carried forward.** Where a number circulates widely but has no traceable primary source, it appears only in §11 (Open questions) or is marked **[U]**, never in a specification table.
2. **Where sources conflict, both are shown and the conflict is named.** See §4.6 and §5.1 for the two live conflicts in the public record.

Companion documents in this library: Intel Arc Pro B70 Reference · Intel Crescent Island (Xe3P) Reference · SambaNova RDU Platform Reference · Memory Technology Reference · Agentic Stack Storage Reference · Xeon 6 Workload/SKU Workbook.

1. Executive summary

The Arc Pro B60 is the full-die Battlemage professional part: 20 Xe2-HPG cores, 160 XMX matrix engines, 24 GB of GDDR6 on a 192-bit bus, and a PCIe 5.0 x8 electrical host link. Intel positions it explicitly as an inference product rather than a rendering product — the data sheet headline is memory capacity and multi-GPU Linux serving, not FP32 throughput. **[I]**

Four properties determine where it fits in an agentic or GenAI serving estate:

- **Capacity per dollar, not FLOPS per dollar.** 24 GB at a street price in the \$599–\$800 band **[3P]** gives roughly one-quarter the memory bandwidth of a contemporary HBM part at a small fraction of the cost. The B60 wins on *how large a model fits* and loses on *how fast a large batch runs*.
- **Arithmetic intensity is high relative to bandwidth.** At 215.6 FLOP/byte for FP16 XMX **[D]**, decode-phase LLM serving on this part is firmly memory-bound; the XMX array will idle unless batch size or prefill length is pushed up. See §4.5.
- **There is no scale-up fabric.** Every inter-GPU byte crosses PCIe 5.0 x8 (~31.5 GB/s per direction **[D]**), a 14.5:1 step down from local GDDR6 bandwidth. Tensor parallelism across cards is therefore expensive; pipeline parallelism and per-card model replication are usually the better shapes. See §5.4.
- **The software stack is the gating item, not the silicon.** Intel's `llm-scaler` containers — not stock vLLM — are the supported serving path, and the release cadence (§6.5) shows the model coverage still moving quickly. Plan for container-version pinning as a first-class operational concern.

The natural deployments are: single-card 8B–14B serving at FP16 or INT4; two-to-four-card 32B–70B serving at INT4/FP8; and eight-card “Project Battlematrix” workstations reaching 192 GB aggregate **[I]** for high-concurrency batch inference where total throughput matters more than per-request latency.

2. Silicon and platform specification

2.1 Intel-published specification

Source: *Intel Arc Pro B60 GPU Data Sheet*, version 1.0. All rows **[I]**.

Category	Parameter	Value
Performance	GPU peak TOPS (INT8, XMX, dense)	197 TOPS
	Peak FP32 throughput	Up to 12.28 TFLOPS
	Xe-cores	20 × Xe2-HPG
	XMX AI engines	160
	Ray tracing units	20
	Host interface	PCIe Gen 5.0 x16 connector, x8 electrical ; Gen 4.0 backward-compatible
Memory	Dedicated memory	24 GB GDDR6
	Bandwidth	456 GB/s
	Interface width	192-bit
Display	Outputs	Varies by board partner
	Max resolution support	2 × 7680×4320 @ 60 Hz; 1 × 5120×1440 @ 240 Hz; 2 × 5120×2880 @ 120 Hz; 4 × 3840×2160 @ 60 Hz
	API support	DirectX 12 Ultimate, oneAPI, OpenCL 3.0, OpenGL 4.6, OpenVINO, Vulkan 1.3
Hardware accel.	XMX AI engines	Yes
	Ray tracing	Yes
	Full encode and decode	AV1, HEVC, H.264, VP9
Power	Total board power	120–200 W (Intel reference specification; partner designs vary)
	Power connector	Varies by board partner
General	Form factor / dimensions	Varies by board partner
	OS support	Microsoft Windows 11 and 10; Linux (Ubuntu)

Intel's own footnotes qualify two of these rows and both qualifications matter for capacity planning:

- The 197 TOPS figure is defined as **peak throughput running XMX workloads with the INT8 datatype on dense models**. It is not an achievable serving number and it does not apply to sparse or MoE-gated computation. **[I]**
- The 12.28 TFLOPS FP32 figure is defined as **maximum clock frequency × peak single-precision operations throughput** — i.e. a boost-clock ceiling, not a sustained rate. **[I]**

2.2 Die and process

Parameter	Value	Tag
GPU die	BMG-G21 (full die, all 20 Xe-cores enabled)	[3P]
Process node	TSMC N5 (5 nm EUV)	[3P]
Transistor count	19.6 billion	[3P]
Die area	272 mm ²	[3P]

Parameter	Value	Tag
Transistor density	≈ 72.1 M/mm ²	[3P]
PCI device ID	0xE211 (B60); 0xE212 observed on a partner SKU	[3P]

The B60 shares BMG-G21 with the consumer Arc B580; the professional part differs in memory capacity (24 GB vs 12 GB on the same 192-bit bus, i.e. 2 GB GDDR6 devices in clamshell), host interface generation (PCIe 5.0 vs 4.0), driver certification, and virtualization enablement. **[3P]**

2.3 Xe2-HPG core organisation

Intel's Xe2 architecture disclosure describes each Xe2 core as: **[II]**

- 8 × 512-bit vector engines (XVE), SIMD16-native, supporting SIMD16 and SIMD32 operations
- 8 × 2048-bit XMX matrix engines
- 64-bit atomic operation support
- 256 KB shared L1 / SLM per Xe-core (up from 192 KB on Alchemist)
- 3-way co-issue: FP + INT/extended-math + XMX in the same clock
- One ray tracing unit per Xe-core

Structurally the die is 5 render slices × 4 Xe-cores = 20 Xe-cores, giving 20 × 8 = 160 XMX engines and 20 × 8 × 16 = 2,560 FP32 lanes. **[D]** The shared L2 is up to 18 MB on BMG-G21. **[3P]**

2.4 Derived clock frequency

Intel does not publish a clock for the B60 reference specification. It is recoverable from the published FP32 figure: **[D]**

$$\begin{aligned} \text{FP32 FLOP/clock} &= 20 \text{ Xe-cores} \times 8 \text{ XVE} \times 16 \text{ lanes} \times 2 \text{ (FMA)} = 5,120 \\ f &= 12.28\text{e}12 / 5,120 = 2,398 \text{ MHz} \end{aligned}$$

This lands on **2,400 MHz**, which is exactly the GPU clock published by ASRock and MAXSUN for their B60 boards **[3P]**. The agreement between an Intel-published throughput figure and an independently-published partner clock is what makes the rest of §5 safe to derive: **2.4 GHz is treated as the reference clock throughout this document.**

2.5 Board partner variants

Intel ships no retail reference card; every B60 is a partner board. Partners named in the public record include ASRock, Gunnir, Lanner, MAXSUN, Onix, Senao and Sparkle. **[3P]**

Variant class	GPUs	Memory	Cooling	Slots	Power	Host link
Creator / Blower 24 GB (e.g. ASRock Creator, Sparkle Blower)	1	24 GB	Active blower	2	1 × 8-pin	PCIe 5.0 x8
Passive 24 GB (e.g. ASRock Passive, Sparkle Passive)	1	24 GB	Passive / chassis airflow	1-2	1 × 8-pin	PCIe 5.0 x8
Dual-GPU 48 GB (e.g. MAXSUN Arc Pro B60 Dual 48G Turbo)	2	2 × 24 GB	Blower	2	1 × 12V-2×6	2 × PCIe 5.0 x8 via x16 bifurcation

All rows **[3P]**, from partner data sheets.

Two board-level facts have direct rack-integration consequences:

- **The dual-GPU 48 GB card is not a single logical GPU.** Each die enumerates as a discrete PCI device. There is no on-board bridge presenting a unified address space, so the host motherboard **must** support x8/x8 bifurcation of the x16 slot. **[3P]**

- **The Sparkle passive 24 GB variant is specified at 200 W TBP** with no onboard fan **[3P]** — a chassis-airflow dependency that must be validated against the server’s thermal design, not assumed.

Display outputs vary: ASRock’s Creator 24 GB carries 4 × DisplayPort 2.1 **[3P]**; the MAXSUN dual card carries 1 × DP 2.1 UHBR20 + 1 × HDMI 2.1a per GPU, with only one of the two outputs per GPU usable at a time **[3P]**.

3. Memory subsystem

3.1 Specification and bandwidth verification

Parameter	Value	Tag
Capacity	24 GB GDDR6	[I]
Bus width	192-bit	[I]
Peak bandwidth	456 GB/s	[I]
Data rate	19 Gbps per pin	[3P]
L2 cache	up to 18 MB (BMG-G21)	[3P]
L1 / SLM	256 KB per Xe-core (5.0 MB aggregate)	[I] / [D]
ECC	Not stated on the B60 data sheet. Listed as a Project Battlematrix platform feature.	[U]

Bandwidth cross-check: [D]

```
192 bits x 19e9 transfers/s / 8 = 456.0 GB/s    -- matches Intel exactly
```

The 19 Gbps partner figure is therefore confirmed by Intel’s own bandwidth number. Note that this is GDDR6, not GDDR6X and not HBM: 456 GB/s is roughly one-eighth of a contemporary HBM3E accelerator. Capacity, not bandwidth, is this part’s differentiator.

On ECC: Intel’s Project Battlematrix material lists ECC among the platform’s enterprise reliability features alongside SR-IOV, telemetry and remote firmware update [3P]. The B60 data sheet itself does not state ECC support, and no Intel document reviewed for this reference confirms whether ECC is on-die, soft/inline, partner-dependent, or capacity-reducing. **Treat ECC as unverified for procurement purposes and confirm with the board partner.** [U]

3.2 Usable capacity budget

Total capacity is not servable capacity. A working budget: [D]

Consumer	Reservation	Basis
Driver, firmware, context state	~0.5 GB	assumption
Runtime + framework working set (oneAPI / PyTorch XPU / vLLM)	~1.0 GB	assumption
Activation and workspace scratch (batch-dependent)	~0.5 GB	assumption
Usable for weights + KV cache	≈ 22 GB	24 – 2

The 22 GB figure is an **assumption block**, not a measurement, and should be replaced with a measured value from `xpu-smi` on the target container image before it is used for procurement. Everything downstream in §3.3 and §3.4 scales linearly with it.

3.3 Model weight footprint

Bytes per parameter by format, and resulting GPU count at 22 GB usable: [D]

Model scale	FP16 (2 B/p)	FP8 (1 B/p)	INT4 g128 (≈0.56 B/p)	GPUs @ INT4	GPUs @ FP16
8 B	16.0 GB	8.0 GB	4.5 GB	1	1
14 B	28.0 GB	14.0 GB	7.9 GB	1	2
32 B	64.0 GB	32.0 GB	18.0 GB	1 (tight)	3

Model scale	FP16 (2 B/p)	FP8 (1 B/p)	INT4 g128 (≈ 0.56 B/p)	GPUs @ INT4	GPUs @ FP16
70 B	140.0 GB	70.0 GB	39.4 GB	2	7
~ 117 B MoE (MXFP4)	—	—	≈ 61 GB	3	—
235 B MoE	470 GB	235 GB	132 GB	6	—

The INT4 column uses 4.5 effective bits/parameter to account for group-wise scales and zero-points at group size 128. GPTQ, AWQ and MXFP4 all land near this figure; per-block FP8 lands near 1.06 B/p. Verify against the actual checkpoint size rather than the nominal parameter count.

The 32 B INT4 single-card row is the trap. 18 GB of weights inside a 22 GB budget leaves ~ 4 GB for KV cache — enough for a handful of short sequences and nothing more. Single-card 32 B is a demo configuration, not a serving configuration.

3.4 KV cache arithmetic

```
bytes_per_token = 2 x n_layers x n_kv_heads x head_dim x bytes_per_element
KV_total        = bytes_per_token x context_length x concurrent_sequences
```

Worked for three grouped-query architectures: **[D]**

Model	Layers	KV heads	Head dim	FP16 B/token	FP8 B/token
Llama 3.1 8B	32	8	128	64 KiB	32 KiB
Qwen3 32B	64	8	128	128 KiB	64 KiB
Llama 3.1 70B	80	8	128	160 KiB	80 KiB

Concurrency envelope for the single-card 8B FP16 case — 22 GB usable minus 16 GB weights = 6 GB KV budget: **[D]**

```
6 GiB / 64 KiB per token = 98,304 tokens of KV headroom
@ 4K context -> 24 concurrent sequences
@ 8K context -> 12 concurrent sequences
@ 32K context -> 3 concurrent sequences
```

Halving to FP8 KV doubles every figure above. Intel's `llm-scaler-vllm` enabled FP8 KV cache in its June 2026 release **[3P]** — on a 24 GB part this is not an optimisation, it is a prerequisite for any long-context serving.

Sizing rule: on the B60, KV cache capacity — not compute — sets the concurrency ceiling in every configuration above 8B FP16. Size the estate on `concurrent_sequences x context_length`, then check compute, never the reverse.

4. Compute throughput by datatype

4.1 Datatype support matrix

Xe2 XMX supports INT2, INT4, INT8, FP16 and BF16 **[I]**. The vector engines additionally handle FP32, FP64 and extended math (SIN, COS, LOG, EXP) **[I]**. Per-clock rates published by Intel for the Xe2 core: **[I]**

Engine	Datatype	Ops per clock per Xe-core
XMX	INT8	4,096
XMX	FP16	2,048

4.2 Derivation model

With 20 Xe-cores at the 2.4 GHz reference clock established in §2.4: **[D]**

$$\text{throughput} = 20 \text{ Xe-cores} \times \text{ops_per_clock_per_core} \times 2.4\text{e9}$$

Validation of the model against Intel’s own number:

INT8: $20 \times 4,096 \times 2.4\text{e9} = 196.6 \text{ TOPS}$ vs Intel's published 197 TOPS

Agreement to within 0.2%. The derivation model is therefore sound, and the FP16 figure below inherits that confidence.

4.3 Throughput table

Datatype	Engine	Peak throughput	Tag	Basis
INT8 (dense)	XMX	197 TOPS	[I]	Intel data sheet
FP16	XMX	98.3 TFLOPS	[D]	$20 \times 2,048 \times 2.4 \text{ GHz}$
BF16	XMX	≈ 98.3 TFLOPS	[D]†	same array, rate assumed equal to FP16
FP32	XVE	12.28 TFLOPS	[I]	Intel data sheet
INT4	XMX	supported; rate not published	[U]	—
INT2	XMX	supported; rate not published	[U]	—
FP16 (vector)	XVE	rate not published	[U]	—
FP64	XVE	supported; rate not published	[U]	—
FP8	—	no native XMX datatype	[I]	absent from Intel’s Xe2 XMX datatype list
TF32	XMX	contested — see §4.6	[3P]	—

† Intel publishes a single XMX rate for FP16 and lists BF16 as a supported XMX datatype but does not publish a separate BF16 rate. Alchemist’s DPAS executed BF16 at the FP16 rate; the same is assumed here. Flag this assumption if BF16 throughput is load-bearing in a sizing model.

4.4 The FP8 nuance — important

Intel’s `llm-scanner-vllm` advertises **dynamic online FP8** quantization for most supported models **[3P]**, and MXFP4 for `gpt-oss` **[3P]**. This does **not** mean the XMX array executes FP8 natively — FP8 does not appear in Intel’s Xe2 XMX datatype list **[I]**.

The correct reading: **FP8 and MXFP4 on this part are weight-storage formats, not compute formats.** They halve or quarter weight footprint and therefore the bytes moved per decode step — which is the binding constraint (§4.5) — but the GEMM still executes at FP16 XMX rates after upconversion. Expect the memory-capacity and memory-bandwidth benefit, do **not** model a compute speedup.

This distinction is the single most common modelling error on Battlemage inference and is worth carrying into the capacity planning application as an explicit flag on the quantization parameter.

4.5 Arithmetic intensity and the roofline

Machine balance — peak compute divided by peak memory bandwidth: **[D]**

Datatype	Peak	Machine balance (ops/byte)
INT8 XMX	197 TOPS	431.2 OP/byte
FP16 XMX	98.3 TFLOPS	215.6 FLOP/byte
FP32 XVE	12.28 TFLOPS	26.9 FLOP/byte

A GEMM must exceed these intensities to be compute-bound. LLM **decode** at batch size 1 has an arithmetic intensity near 2 FLOP/byte — two orders of magnitude below the FP16 balance point. **Single-stream decode on the B60 uses well under 1% of the XMX array.**

First-order decode ceiling, at an assumed 70% achieved bandwidth: **[D]**

```
tokens/s ~= (456 GB/s x 0.70) / weight_bytes
```

Model / format	Weight bytes	Ceiling, batch 1
8B INT4	4.5 GB	≈ 71 tok/s
8B FP16	16.0 GB	≈ 20 tok/s
32B INT4	18.0 GB	≈ 18 tok/s

The 70% efficiency term is an assumption **[D]**; substitute a measured value. These are ceilings under ideal conditions, before attention, sampling and framework overhead.

Planning consequences. Batching is not an optimisation on this part, it is the whole design. Prefill is compute-bound and will use the XMX array; decode is bandwidth-bound and will not. If the workload is agentic — many short, latency-sensitive, low-concurrency steps — the B60 will spend most of its life bandwidth-starved and the FLOPS number on the data sheet is close to irrelevant. Prefix caching and FP8 KV are worth more than any compute tuning.

4.6 Contested: TF32

Two reviewer write-ups of Intel's Xe2 architecture briefing list **TF32** among XMX-supported datatypes **[3P]**. Several other transcriptions of the same briefing list only INT2, INT4, INT8, FP16 and BF16 **[3P]**, and Intel's own Xe-HPG developer article gives FP16, BF16, INT8, INT4, INT2 for the predecessor **[I]**.

No Intel document reviewed for this reference confirms TF32 on Xe2 XMX. It is recorded here as **contested and not to be relied upon**. See §11.

5. Host interface and multi-GPU scaling

5.1 PCIe budget

Parameter	Value	Tag
Connector	PCIe 5.0 x16 mechanical	[I]
Electrical	x8	[I]
Backward compatibility	PCIe 4.0	[I]

Bandwidth: **[D]**

```
PCIe 5.0 = 32 GT/s per lane, 128b/130b encoding
x8 per direction = 32e9 x 8 x (128/130) / 8 = 31.5 GB/s
aggregate bidirectional = 63.0 GB/s
```

Ratio to local GDDR6: **456 / 31.5 = 14.5:1**. **[D]** Any tensor that has to leave the card costs 14.5× what it costs to read locally.

Conflict in the public record. At least one published preview states 128 GB/s bidirectional for a PCIe 5.0 x8 link, described as matching PCIe 4.0 x16 **[3P]**. This is incorrect on both counts: PCIe 4.0 x16 is 32 GB/s per direction, and PCIe 5.0 x8 is 31.5 GB/s per direction. The derived figures above are the ones to plan against.

5.2 Lane budget for a dense host

Each B60 die needs 8 PCIe 5.0 lanes; each dual-GPU 48 GB card needs a bifurcated x16. An eight-GPU Battlematrix configuration therefore consumes **64 PCIe 5.0 lanes** for GPUs alone **[D]**, before NICs, NVMe and any management devices. On a single-socket Xeon 6 host this is the dominant lane consumer and must be reconciled against the platform's total lane budget and root-port bifurcation capability at design time, not at rack time.

The motherboard must support **x8/x8 bifurcation** on every slot carrying a dual card. This is a firmware capability, not a physical one — validate it on the specific BIOS version.

5.3 Project Battlematrix

Intel's reference configuration: up to **8 × Arc Pro B60** in a workstation chassis for **192 GB** aggregate GDDR6, targeting 70B-plus parameter models, paired with a Xeon host and a validated containerized Linux stack. Announced system pricing was quoted in the \$5,000–\$10,000 band. **[3P]**

5.4 The absent scale-up fabric

There is no NVLink-class, UALink-class or vendor-proprietary GPU-to-GPU interconnect on the B60. `llm-scaler-vllm` implements collectives via oneCCL with P2P or USM transport **[3P]** — but that traffic still crosses PCIe, and on many platforms peer-to-peer between root ports degrades to a host-bounce path.

Design consequences, in priority order:

1. **Prefer replication over tensor parallelism.** N independent single-card replicas behind a load balancer will out-serve an N-way tensor-parallel deployment of the same model whenever the model fits on one card. Tensor parallelism puts an all-reduce on the critical path of every layer, over a link 14.5× slower than local memory.
2. **When the model does not fit, prefer pipeline parallelism.** It moves activations at layer boundaries rather than partial sums at every layer — far less PCIe traffic per token. `llm-scaler-vllm` supports tensor, pipeline and data parallelism **[3P]**.

3. **Do not plan disaggregated prefill/decode across B60 cards.** KV transfer between a prefill and a decode pool is a bulk-bytes operation sized for a scale-out fabric; 31.5 GB/s per card will not carry it.
 4. **Keep the eight-card configuration for throughput, not latency.** Reviewer testing found the full eight-GPU configuration advantageous for batch inference, synthetic data generation and high-concurrency serving, while low-concurrency single-user workloads saw acceptable performance from much smaller configurations **[3P]**.
-

6. Software stack

6.1 Layer map

Layer	Component	Notes	Tag
Kernel	Intel <code>xe</code> DRM driver (upstream Linux)	SR-IOV for Battlemage upstreamed in Linux 6.17	[3P]
Firmware	GPU firmware via <code>igsc</code> ; OPROM	SR-IOV exposure is firmware-gated — see §6.6	[3P]
User-mode runtime	Intel Compute Runtime (NEO), Level Zero		[3P]
Programming model	oneAPI / SYCL (DPC++), OpenCL 3.0	OpenCL and oneAPI listed on the data sheet	[I]
Libraries	oneDNN, oneCCL, oneMKL	oneCCL provides the P2P/USM collectives	[3P]
Framework	PyTorch XPU backend	PyTorch 2.9 → 2.10 across 2026 container releases	[3P]
Inference toolkit	OpenVINO	Listed on the data sheet	[I]
Serving	<code>intel/llm-scaler-vllm</code> (vLLM fork)	The supported LLM serving path	[3P]
Multimodal	<code>intel/llm-scaler-omni</code> (ComfyUI, SGLang Diffusion, X inference)	Image / video / audio / 3D generation	[3P]
Graphics APIs	DirectX 12 Ultimate, Vulkan 1.3, OpenGL 4.6		[I]

6.2 Windows and ISV certification

Windows 11 and 10 are supported **[I]**. Intel differentiates the professional driver from the standard Arc driver by ISV certification for AI, design, engineering, architectural and manufacturing applications, and states that certification and driver availability **vary by board partner** — it is not a property of the silicon **[I]**. If ISV certification is part of the procurement requirement, it must be validated against the specific partner SKU.

6.3 LLM Scaler — the serving path

`llm-scaler` is Intel's GenAI solution for Arc Pro B60 and B70, built on standard frameworks (vLLM, ComfyUI, SGLang Diffusion, X inference) and distributed as Docker images with bare-metal deployment support **[3P]**.

`llm-scaler-vllm` features **[3P]**:

- oneCCL support (P2P or USM transport)
- INT4 and FP8 quantized online serving; MXFP4 for `gpt-oss`
- Embedding and reranker model support
- Multi-modal and omni model support
- Tensor, pipeline and data parallelism
- Maximum-context-length discovery utility
- Multi-modal WebUI; BPE-Qwen tokenizer
- OpenAI-compatible endpoints via `vllm.entrypoints.openai.api_server`

6.4 Model coverage

Model families validated in the `llm-scaler-vllm` support matrix, by class **[3P]**:

Class	Families
Dense text	Llama 3.1 (8B/70B), Qwen3 (8B/14B/32B), Qwen3.5-27B, QwQ-32B, DeepSeek-R1 distills (1.5B–70B), Ministral-8B, Seed-OSS-36B, Hunyuan, Baichuan2, CodeGeeX4, GLM-4
MoE	Qwen3-30B-A3B, Qwen3-235B-A22B, Qwen3.5-35B-A3B / 122B-A10B, Mixtral-8x7B, GLM-4.5-Air, gpt-oss-20b / 120b (MXFP4)
Vision-language	Qwen2.5-VL (7B/32B/72B), Qwen3-VL, InternVL2/3, MiniCPM-V, Gemma 3 (12B/27B), GLM-4.1V, Kimi-VL, ERNIE-4.5-VL
OCR / document	MinerU 2.5, PaddleOCR-VL, DeepSeek-OCR, dots.ocr
Audio / omni	Whisper (medium, large-v3), Qwen2.5-Omni-7B, Qwen3-Omni-30B-A3B
Embedding / rerank	Qwen3-Embedding-8B, Qwen3-Reranker-8B, BGE-M3, BGE-reranker-v2-m3, Qwen3-VL-Embedding/Reranker
Diffusion / video (omni)	Stable Diffusion 3.5, Flux.1 / Kontext, Qwen-Image, Z-Image-Turbo, Wan2.2 (T2V/I2V/Animate), HunyuanVideo 1.5, LTX-2, Hunyuan3D 2.1, SeedVR2

Coverage in the FP16 / online-FP8 / online-INT4 / MXFP4 columns is model-specific and several entries carry operational caveats (forced dtype, required chat template, disabled MLA, pinned `transformers` version). **Consult the support matrix for the exact model before committing to a deployment shape.**

6.5 Release cadence

The container release history is the clearest available signal on stack maturity **[3P]**:

Date	Release	Significance
2025.08	<code>llm-scaler-vllm:1.0</code>	First stable container. Claimed up to 1.8× TPOP uplift on 32B at >4K input and 4.2× on 70B at 40K sequence length.
2025.09	<code>vllm:0.10.0-b3</code> , <code>omni:0.1.0-b1</code>	First multimodal/omni enablement on B60.
2025.11	<code>vllm:0.10.2-b5</code> / <code>-b6</code>	<code>gpt-oss</code> , Qwen3-VL, Qwen3-Omni, MinerU 2.5. Windows install for omni.
2026.01	<code>vllm:1.3</code> (<code>0.11.1-b7</code>)	vLLM 0.11.1, PyTorch 2.9.
2026.03	<code>vllm:0.14.0-b8</code> / <code>-b8.1</code>	vLLM 0.14.0, PyTorch 2.10; Qwen3.5 family.
2026.05	<code>vllm:1.4</code> (<code>0.14.0-b8.2.1</code>)	New platform image; Arc Pro B70 support added .
2026.06	<code>vllm:0.14.0-b8.3.1</code> / <code>.2</code>	FP8 KV cache enabled ; Qwen3.5/3.6 accuracy fixes.
2026.07	<code>vllm:0.21.0-b1</code> , <code>omni:0.1.0-b8</code>	Gemma-4 family; experimental XPU graph; ComfyUI 0.27.0.
2026.08	<code>vllm:0.21.0-b2</code>	Multi-token prediction (MTP); LoRA serving; per-block FP8 for Qwen3.6.

Two operational reads. First, the stack is still moving fast enough that **container version is a load-bearing configuration item** — pin it, test upgrades, and expect model support to be the reason to move rather than performance. Second, the June 2026 FP8 KV cache release and the August 2026 MTP/LoRA release materially change the capacity model for a 24 GB part; any sizing done before those dates understates what the hardware can serve.

6.6 Virtualization and manageability

SR-IOV is enabled for Battlemage **only on Arc Pro parts**, not consumer Arc B-series **[3P]**. Support was upstreamed to the Intel `xe` kernel driver in **Linux 6.17** **[3P]**.

The B60's SR-IOV story required firmware to catch up with the kernel, and this produced visible confusion in the field. The verified position as of early 2026 **[3P]**:

- Kernel 6.17 or newer is required.
- The GPU firmware must be SR-IOV-capable. Cards shipped with earlier firmware do **not** expose the SR-IOV capability in `lspci` regardless of BIOS settings, CPU, or platform.
- Firmware from the February 2026 Arc Pro driver package (reported as 32.0.101.8314) enabled the capability, with **up to 7 virtual functions** observed.
- IOMMU should be enabled; above-4G decoding and resizable BAR are prerequisites.

Do not accept a card's SR-IOV status from `lspci` alone without first confirming the firmware level via `igsc`. Public support threads contain confident but incorrect assertions that the B60 does not support SR-IOV at all; those predate the firmware enablement.

Other Battlematrix platform manageability features named by Intel: ECC (see §3.1 caveat), telemetry monitoring, and remote firmware update **[3P]**.

7. Deployment notes and anti-patterns

#	Anti-pattern	Why it fails	Do instead
1	Sizing from the 197 TOPS headline	It is a dense-INT8 peak; decode is bandwidth-bound at <1% XMX utilisation	Size on KV capacity and 456 GB/s
2	Modelling FP8/MXFP4 as a compute speedup	FP8 is not a native XMX datatype (§4.4)	Model it as a footprint and bandwidth reduction only
3	Tensor-parallel across cards when the model fits on one	All-reduce on a 14.5:1 slower link, every layer	Replicate per card, load-balance in front
4	Assuming x16 bandwidth from an x16 connector	Electrically x8	Budget 31.5 GB/s per direction per die
5	Ordering dual 48 GB cards without checking bifurcation	Two discrete PCI devices need x8/x8 firmware support	Validate bifurcation on the exact BIOS version
6	32 B INT4 on a single card in production	18 GB weights leaves ~4 GB KV — no concurrency headroom	Two cards, or 14 B on one
7	FP16 KV cache on a 24 GB part	Halves the concurrency ceiling for no accuracy benefit at serving scale	Enable FP8 KV (June 2026 container or later)
8	Passive cards in an unvalidated chassis	200 W with no onboard fan is an airflow dependency	Validate chassis thermals against the partner spec
9	Floating the container version	Model support and KV features move release-to-release	Pin the image; gate upgrades on a model-support need
10	Assuming ECC	Not stated on the B60 data sheet (§3.1)	Confirm with the board partner before committing
11	Disaggregated prefill/decode across cards	KV transfer needs a fabric this part does not have	Keep prefill and decode co-resident
12	Assuming SR-IOV works out of the box	Firmware-gated; needs kernel $\geq$ 6.17 and Feb-2026-or-later firmware	Check <code>igsc</code> firmware level, then <code>lspci</code>

8. Positioning within the Arc Pro line

The Arc Pro B-series spans B50, B60, B65 and B70 **[3P]**. The B60 is the full-die Battlemage part at 24 GB; the B70 sits above it and gained `llm-scaler` support in the May 2026 container release **[3P]**. The B50 (16 GB, 70 W, \$299 launch price) is the efficiency-oriented entry point **[3P]**.

For this library, the practical division of labour is:

- **B50** — edge and desktop inference where 16 GB suffices and power budget is the constraint.
 - **B60** — the capacity-per-dollar workhorse; multi-card Linux serving of 8B-70B models. The subject of this document.
 - **B70** — see the companion Arc Pro B70 reference in this library.
 - **Crescent Island (Xe3P)** — the forward-looking, LPDDR5X-based, inference-optimised data centre part; see the companion Crescent Island reference.
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9. Cross-references to the Xeon 6 SKU workbook

Intel_Xeon6_Workload_SKU_Recommendations.xlsx, sheet SKU Recommendations:

Workbook rows	Category / workload	Relevance to a B60 deployment
77-82	AI / GenAI Infrastructure → <i>Inference of Large GenAI models</i> → GPU host CPU	The primary mapping. A B60 host is a GPU host CPU: the SKU choice is driven by PCIe lane count and memory bandwidth for data staging, not by AMX throughput.
83-86	AI / GenAI Infrastructure → <i>Vector Search & Indexing</i>	Retrieval sits on the CPU side of a B60 RAG deployment. Pair with the disk-resident indexing guidance in the storage reference.
71-76	AI / GenAI Infrastructure → <i>Training of Large GenAI models</i> → GPU host CPU	Not applicable. The B60 is an inference part; no training fabric, no scale-up interconnect.
137-139	Storage / DB / VDI / AI (1-socket) → <i>Artificial Intelligence, 1-socket, no discrete GPU</i>	The alternative to a B60 deployment — Xeon 6 AMX-only inference. Compare on tokens/s per rack-watt before choosing.

Host selection note. The lane budget in §5.2 (64 PCIe 5.0 lanes for eight dies) is the binding constraint on host SKU choice, ahead of core count. When reconciling against the workbook's *GPU host CPU* rows, check the platform lane budget and root-port bifurcation capability first; core count second.

10. Open questions register

#	Question	Status	Impact if wrong
1	Does the B60 support ECC, and at what capacity cost?	[U] — Battlematrix platform feature list mentions ECC; B60 data sheet does not	Changes usable capacity and enterprise suitability
2	INT4 / INT2 XMX per-clock rate on Xe2	[U] — datatypes confirmed, rates not published	Would change the INT4 serving compute ceiling
3	Is TF32 an Xe2 XMX datatype?	Contested (§4.6)	Minor; TF32 is not a serving format here
4	BF16 XMX rate — equal to FP16?	[D] , assumed by analogy to Alchemist DPAS	Would halve or double the BF16 ceiling
5	XVE (vector) FP16 and FP64 rates on Xe2	[U]	Affects non-XMX HPC-style kernels only
6	Measured achievable memory bandwidth (% of 456 GB/s)	[U] — 70% assumed in §4.5	Directly scales every decode throughput estimate
7	Measured runtime memory reservation on the target container	[U] — 2 GB assumed in §3.2	Directly scales every capacity and concurrency figure
8	Does oneCCL P2P avoid a host bounce on a given platform?	[U] — platform and root-port dependent	Determines whether tensor parallelism is viable at all
9	Official Intel MSRP	[U] — Intel indicated ~\$500 at announcement; retail observed \$599–\$800	Changes cost-per-GB comparisons
10	Sustained clock under 120 W vs 200 W TBP	[U] — all throughput figures assume the 2.4 GHz reference clock	Low-TBP partner SKUs may not hit derived peaks

Items 6 and 7 are the two highest-leverage measurements to take on first-article hardware. Both are single-command measurements and both scale the entire capacity model linearly.

11. Source register

Intel-published [I]

1. *Intel Arc Pro B60 GPU Data Sheet, v1.0* — download.intel.com/newsroom/2025/client-computing/Intel-Arc-Pro-B60-Data-Sheet.pdf
2. *Intel Arc Pro B60 Datasheet* landing page — intel.com/content/www/us/en/content-details/855693/intel-arc-pro-b60-datasheet.html
3. *Introduction to the Xe-HPG Architecture* (Intel developer article) — predecessor DPAS datatype support
4. Intel VTune Profiler documentation — XVE FP16/BF16/INT8/INT4/INT2 XMX instruction counters
5. *Project Battlematrix: Software Update* (Intel developer article, August 2025)

Third-party [3P]

1. github.com/intel/llm-scaler — README, support matrix, release history (Intel-maintained repository; classified 3P as it is not a product specification document)
2. ASRock product pages — Arc Pro B60 Creator 24 GB, Arc Pro B60 Passive 24 GB
3. Sparkle product pages — Arc Pro B60 24 GB Blower / Passive / Passive 48 GB
4. MAXSUN — Arc Pro B60 Dual 48G Turbo specification page and launch release
5. Phoronix — Arc Pro B-series announcement; SR-IOV restricted to Arc Pro
6. TechPowerUp — Battlemage architecture deep-dive; Project Battlematrix and LLM Scaler 1.0 coverage; B60 retail listing
7. Tom's Hardware — Arc Pro B50/B60 launch; LLM Scaler 1.0; Battlemage architecture
8. Jon Peddie Research — Xe2 core structure and XMX per-clock rates
9. StorageReview — Arc Pro B60 Battlematrix preview (note: PCIe figure disputed, \$5.1)
10. VideoCardz — Arc Pro B60/B50 announcement; retail availability
11. Intel Community forums — SR-IOV firmware enablement threads (Feb 2026)
12. Level1Techs forum — B60 SR-IOV with 7 VFs after `igsc` firmware update
13. Dolphin Interconnect eXpressWare documentation — Arc Pro SR-IOV kernel requirement

Compiled 11 August 2026. Figures tagged **[D]** are derived from tagged **[I]** inputs with the formula shown inline; figures tagged **[U]** are explicitly unpublished and must not be inferred. Board-partner specifications vary — confirm against the specific SKU before procurement.